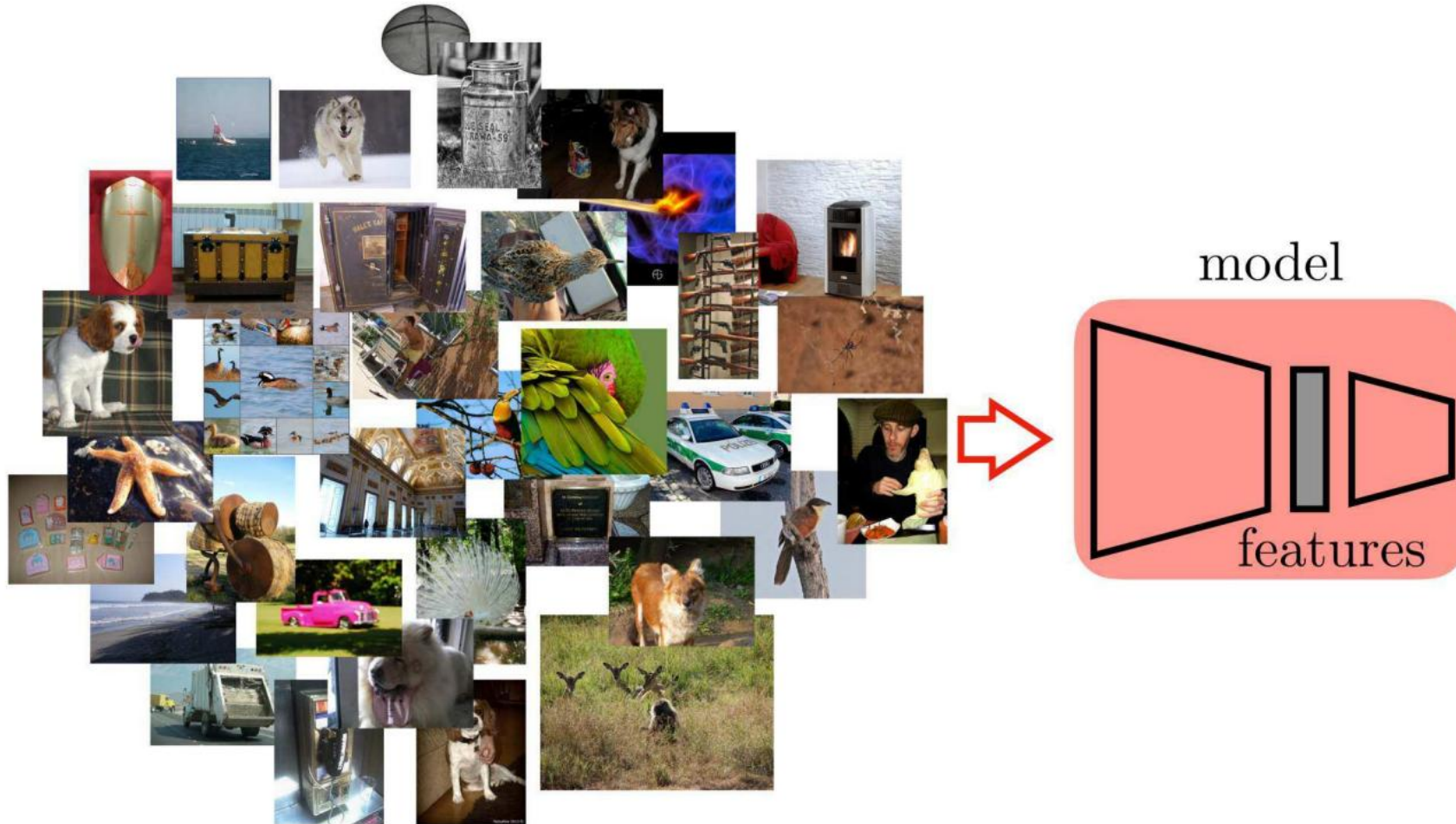


A baby with light brown hair and blue eyes is sitting on a green surface, wearing a blue patterned shirt. The baby is holding two alphabet blocks, one yellow with the letter 's' and one green with the letter 'x'. Other blocks are visible on the surface. The background is a solid green color.

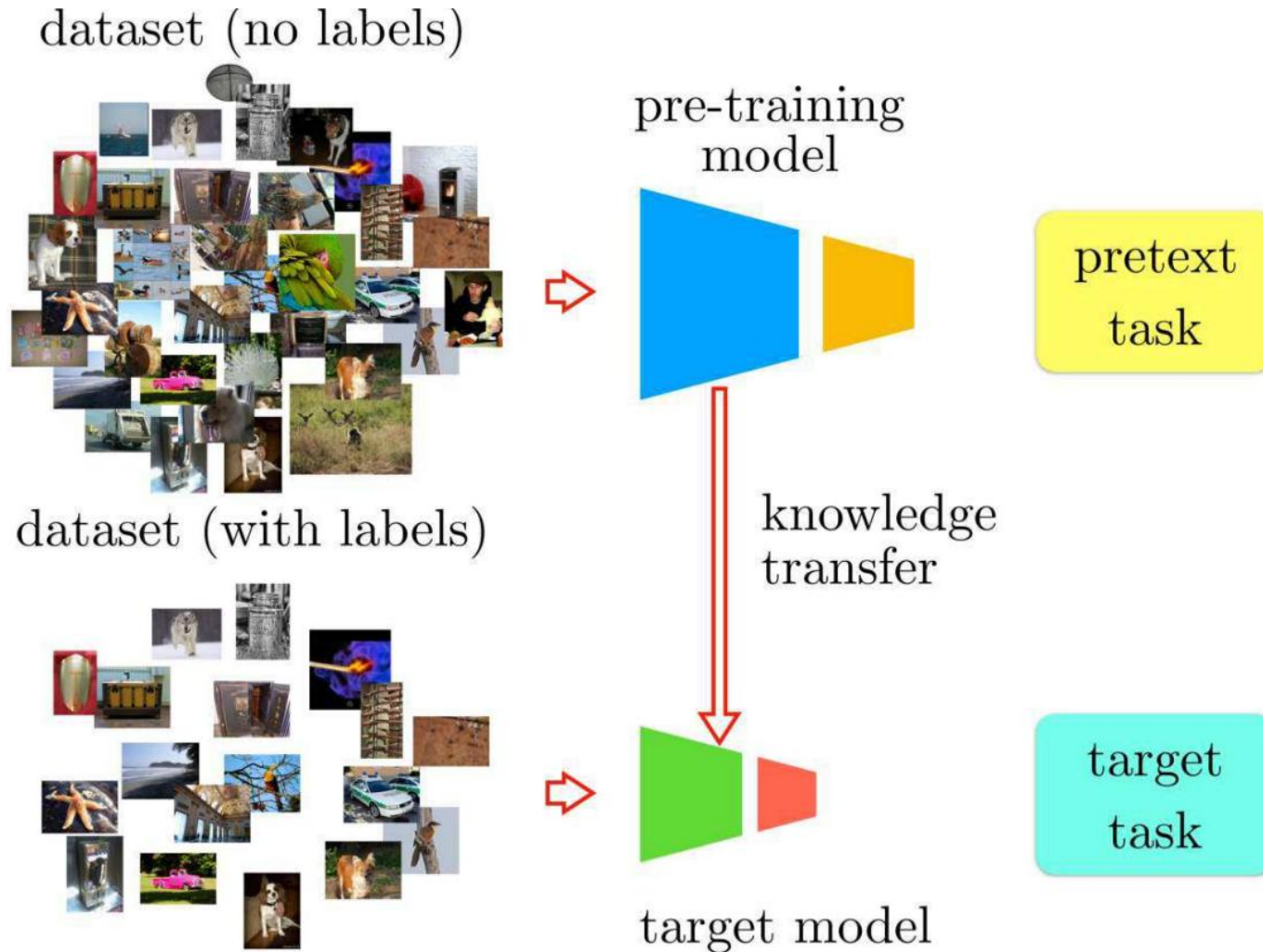
Self-Supervised Learning

Motivation

dataset (no labels)



Motivation



Motivation

Google's medical imaging analysis model

- Multi-Instance Contrastive Learning (MICLe)

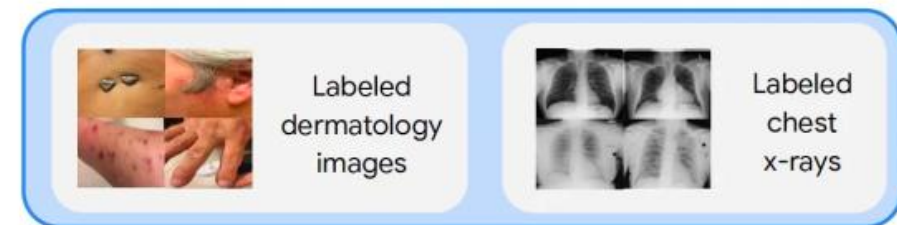
(1) Self-supervised learning on **unlabeled** natural images



(2) Self-supervised learning on **unlabeled** medical images and **Multi-Instance Contrastive Learning (MICLe)** if multiple images of each medical condition are available



(3) Supervised fine-tuning on **labeled** medical images



Pretext tasks

Example: learn to predict image transformations / complete corrupted images

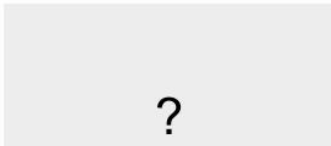
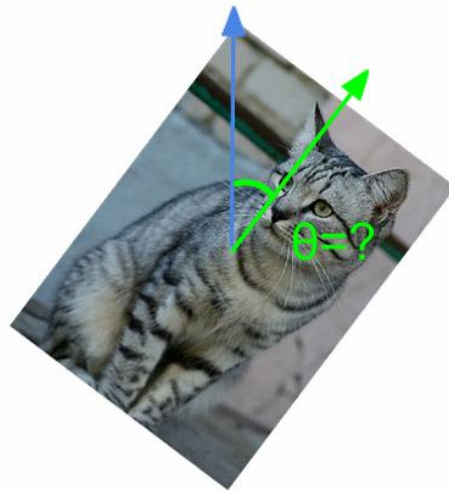


image completion



rotation prediction



“jigsaw puzzle”



colorization

Pretext task: predict rotations



90° rotation



270° rotation



180° rotation



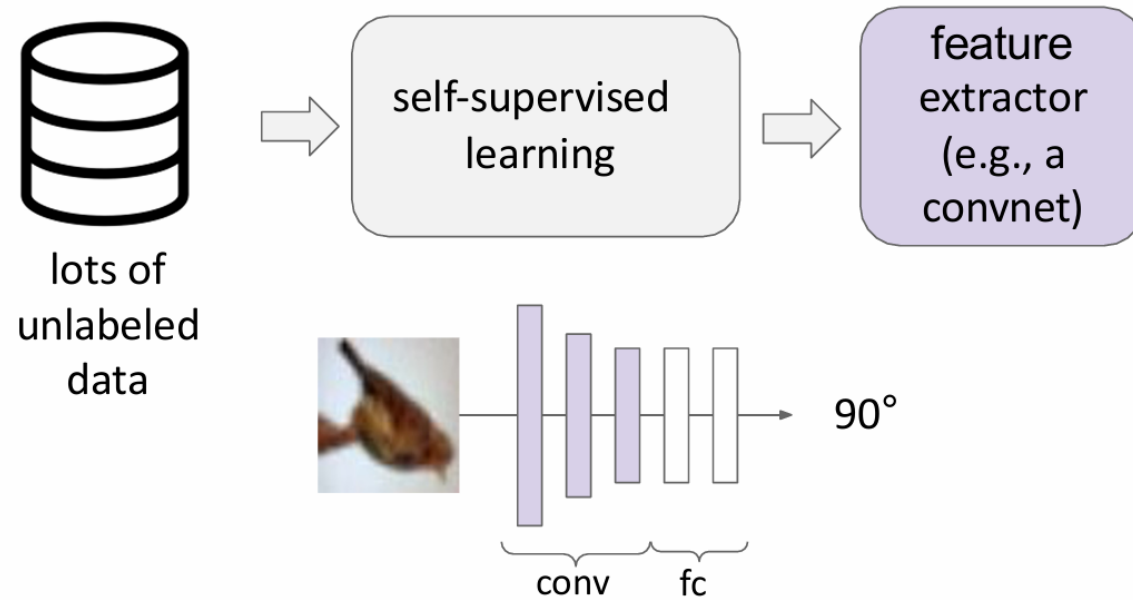
0° rotation



270° rotation

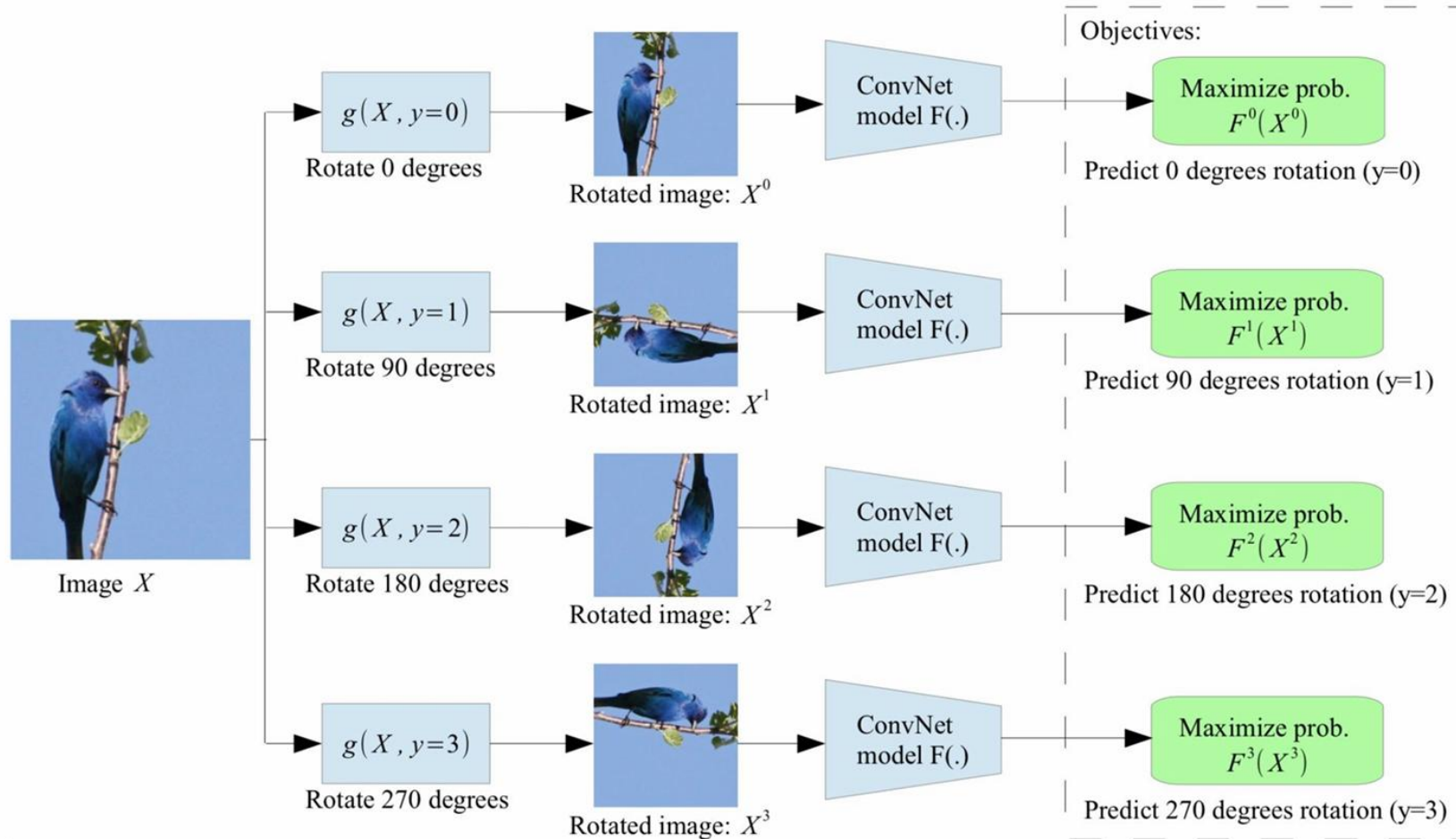
Hypothesis: a model could recognize the correct rotation of an object only if it has the “visual commonsense” of what the object should look like unperturbed.

Pretext task: predict rotations



1. Learn good feature extractors from self-supervised pretext tasks, e.g., predicting image rotations

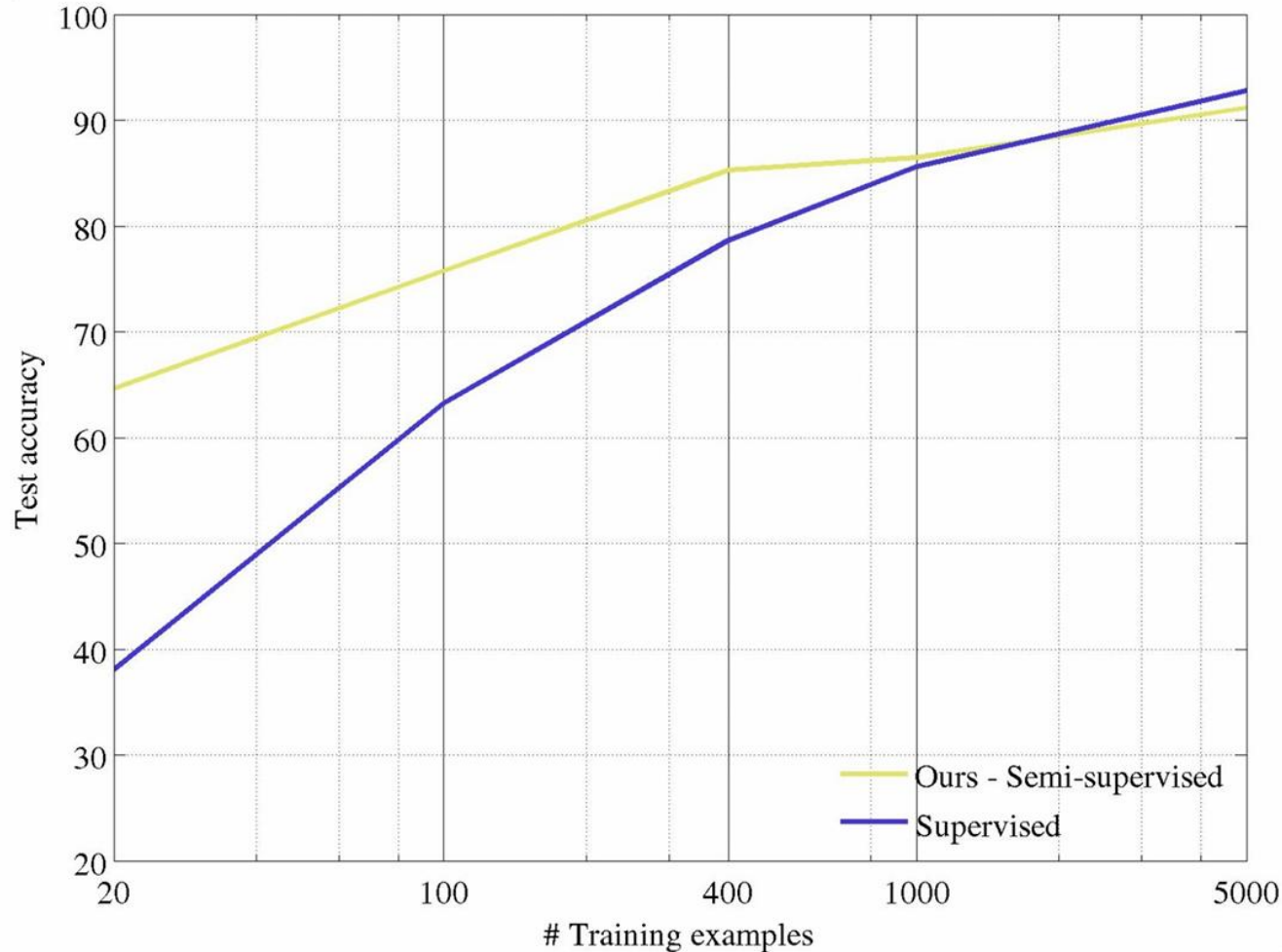
Pretext task: predict rotations



Self-supervised learning by rotating the entire input images.

The model learns to predict which rotation is applied (4-way classification)

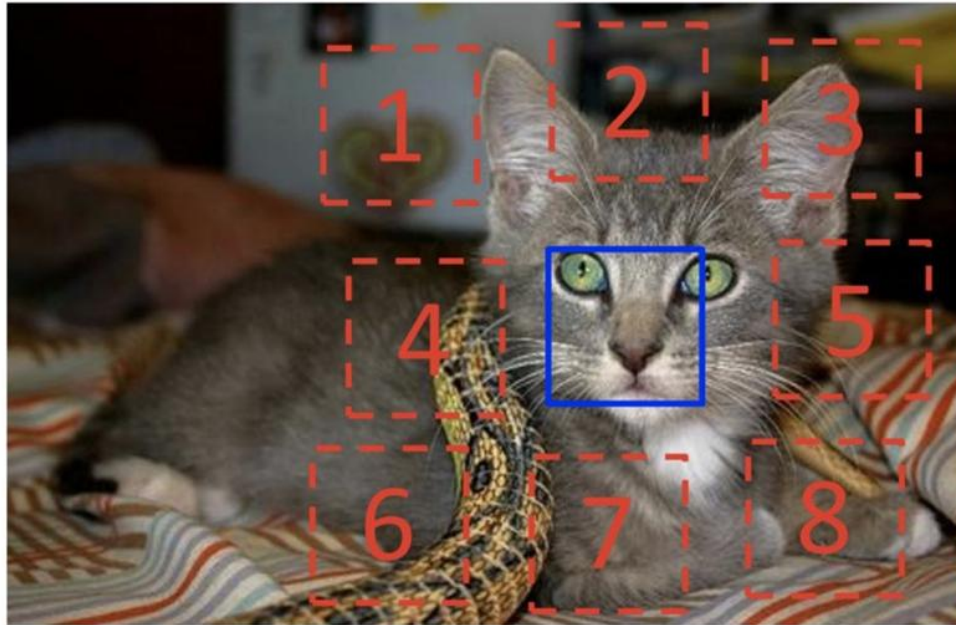
Evaluation on semi-supervised learning



Self-supervised learning on **CIFAR10** (entire training set).

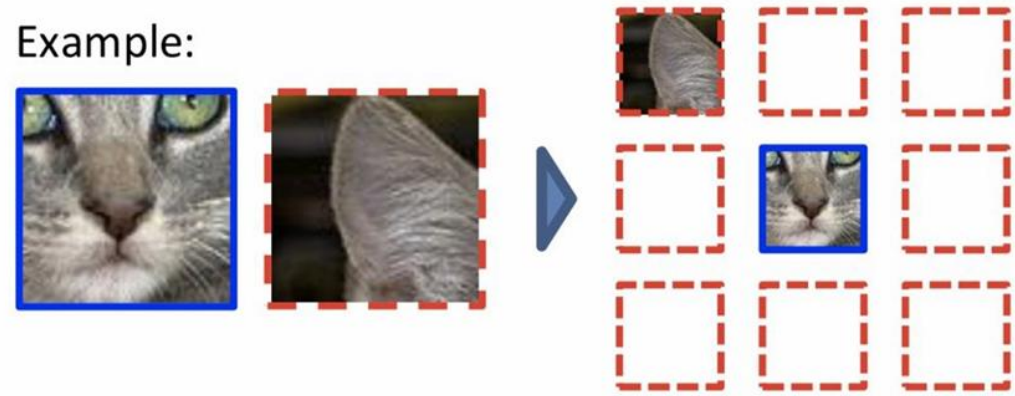
Freeze conv1 + conv2 Learn **conv3 + linear** layers with subset of labeled CIFAR10 data (classification).

Pretext task: predict relative patch locations



$$X = (\text{cat_face}, \text{cat_ear}); Y = 3$$

Example:



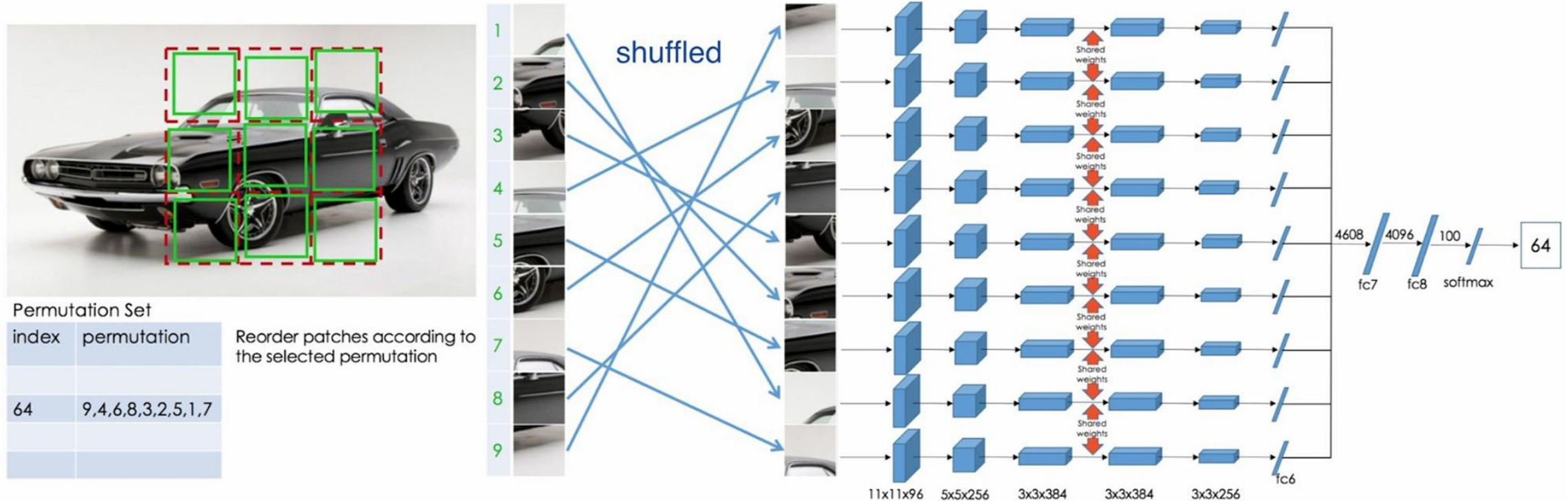
Question 1:



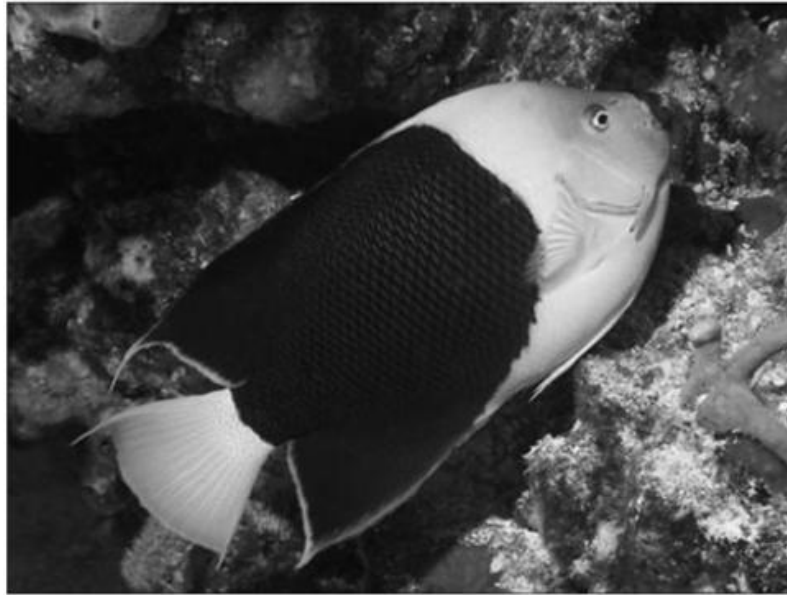
Question 2:



Pretext task: solving “jigsaw puzzles”



Pretext task: image coloring



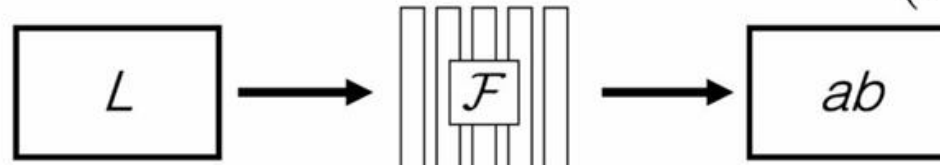
Grayscale image: L channel

$$\mathbf{X} \in \mathbb{R}^{H \times W \times 1}$$



Concatenate (L, ab) channels

$$(\mathbf{X}, \hat{\mathbf{Y}})$$



Pretext task: image coloring

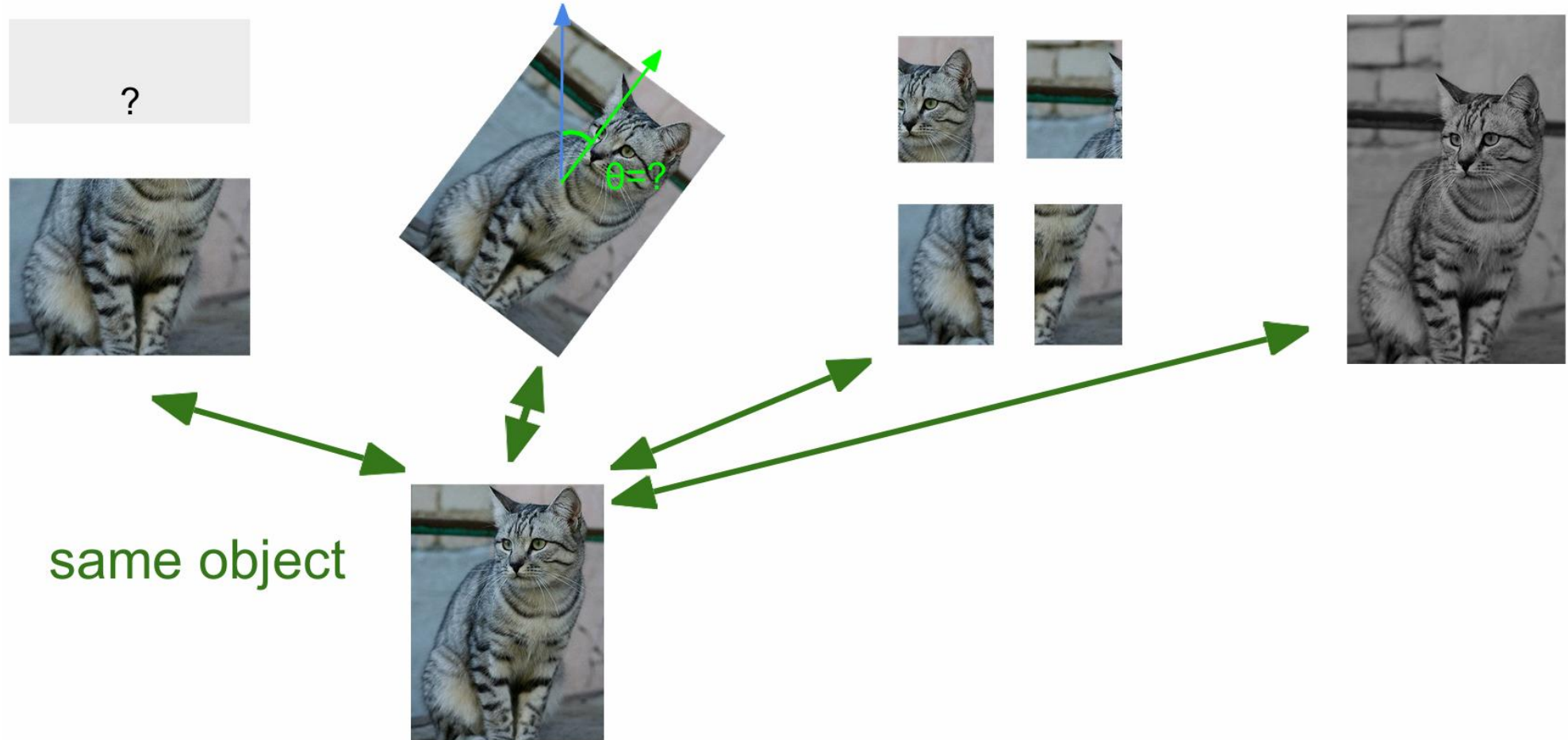


Summary: pretext tasks

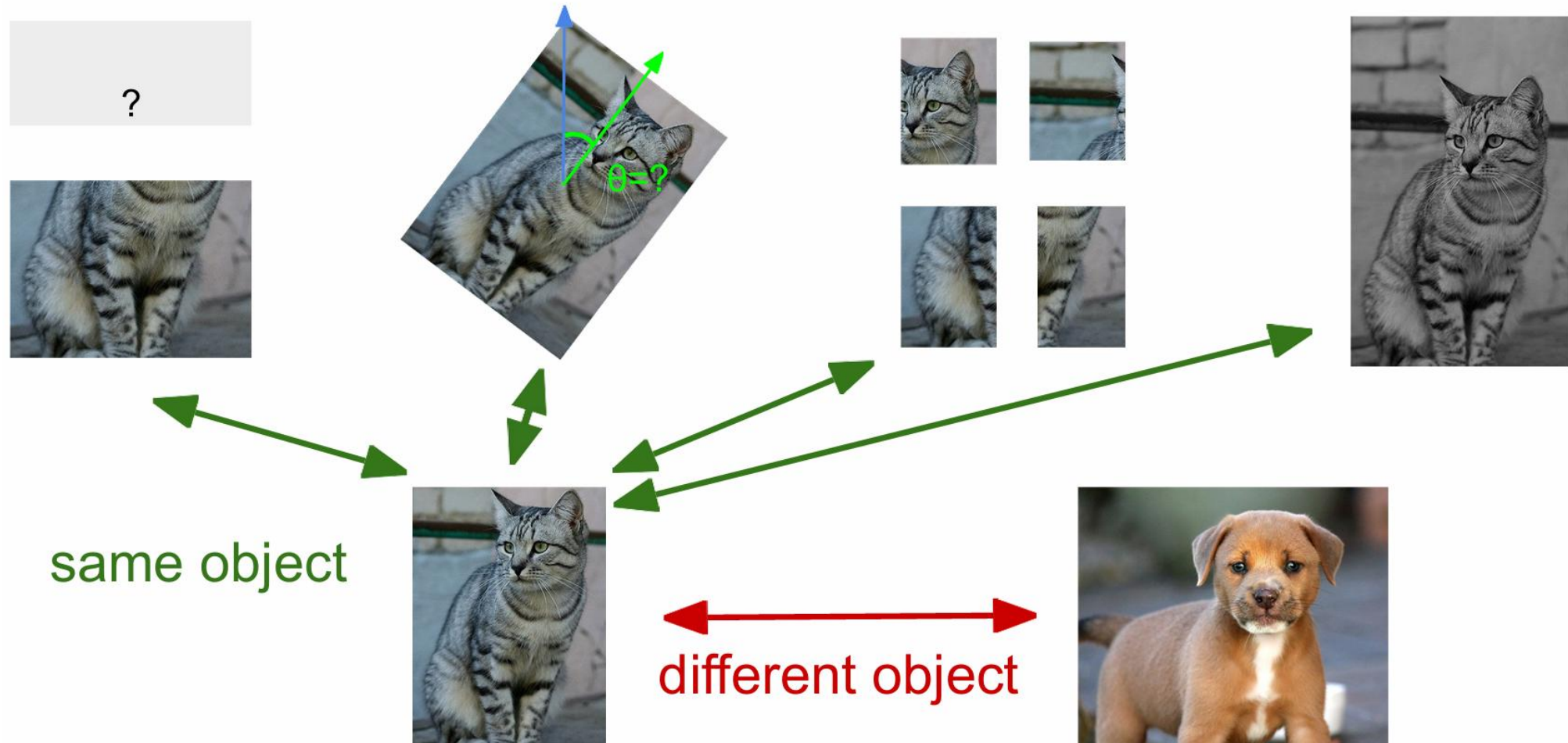
- Pretext tasks focus on “visual common sense”, e.g., predict rotations, inpainting, rearrangement, and colorization.
- The models are forced learn good features about natural images, e.g., semantic representation of an object category, in order to solve the pretext tasks.
- We don’t care about the performance of these pretext tasks, but rather how useful the learned features are for downstream tasks (classification, detection, segmentation).
- **Problems:**
 1. coming up with individual pretext tasks is tedious
 2. the learned representations may not be general.

Learned representations may be tied to a specific pretext task! Can we come up with a more general pretext task?

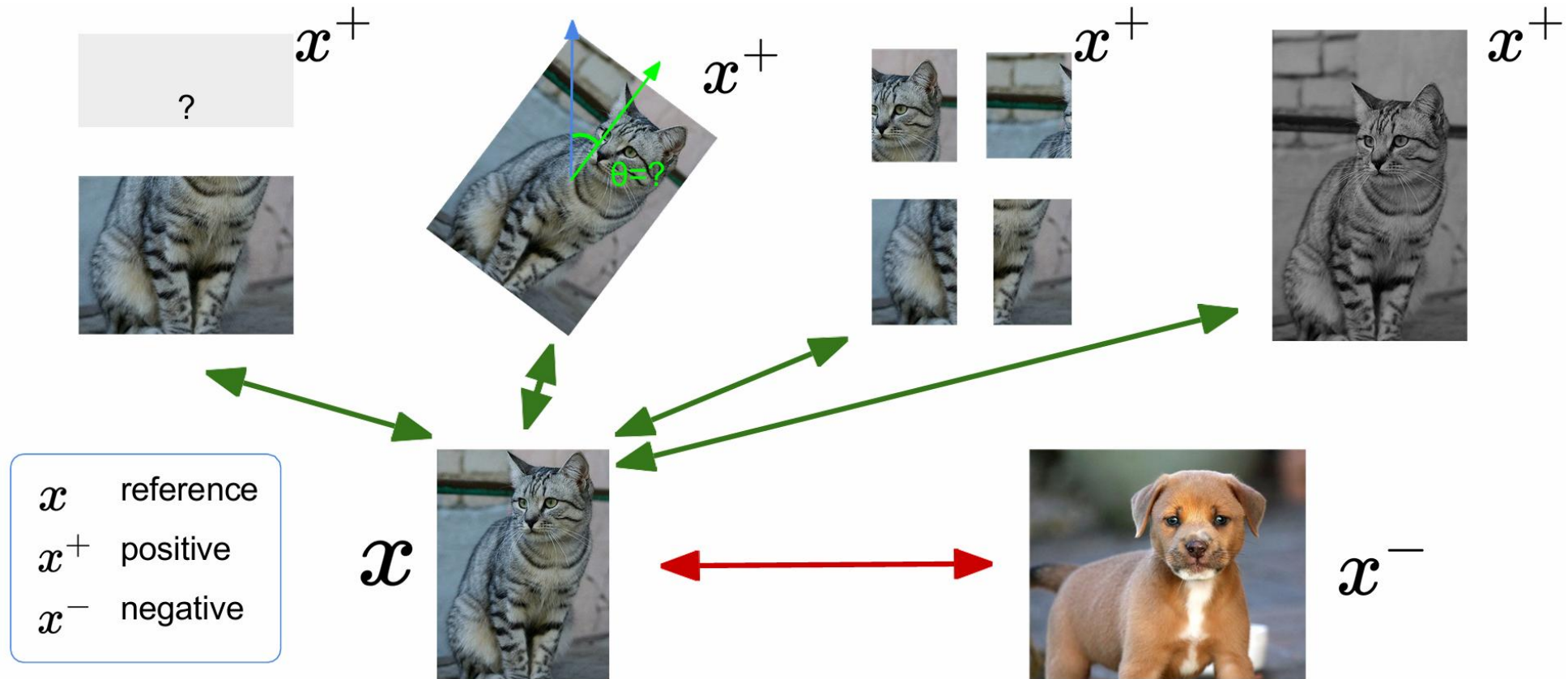
General Pretext Task



General Pretext Task



Contrastive Representation Learning



$$\text{score}(f(x), f(x^+)) \gg \text{score}(f(x), f(x^-))$$

Contrastive learning

Loss function given 1 positive sample and N - 1 negative samples:

$$L = -\mathbb{E}_X \left[\log \frac{\exp(s(f(x), f(x^+)))}{\exp(s(f(x), f(x^+))) + \sum_{j=1}^{N-1} \exp(s(f(x), f(x_j^-)))} \right]$$



x



x^+



x



x_1^-



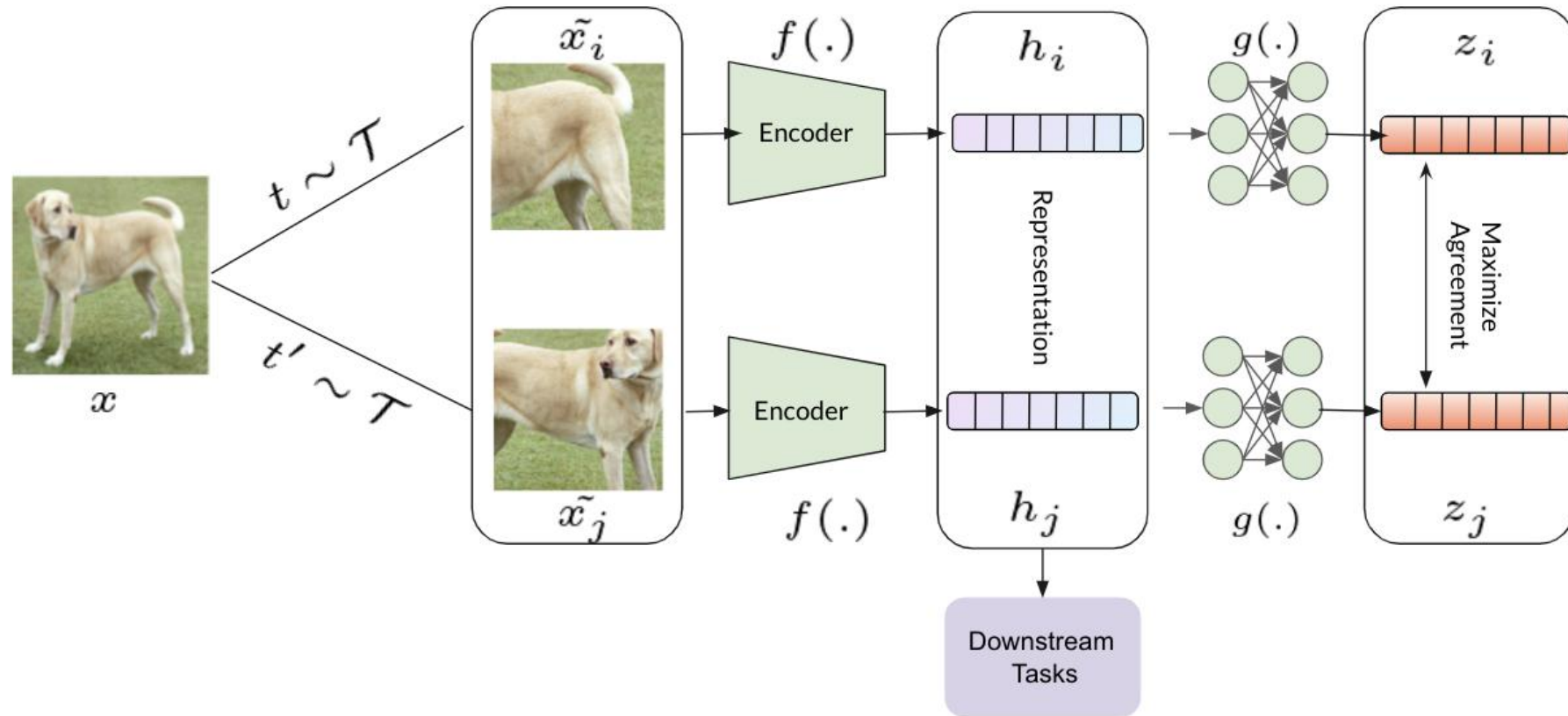
x_2^-



x_3^-

...

SimCLR: A Simple Framework for Contrastive Learning



SimCLR



(a) Original



(b) Crop and resize



(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate $\{90^\circ, 180^\circ, 270^\circ\}$



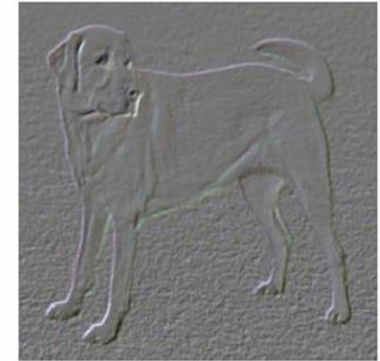
(g) Cutout



(h) Gaussian noise

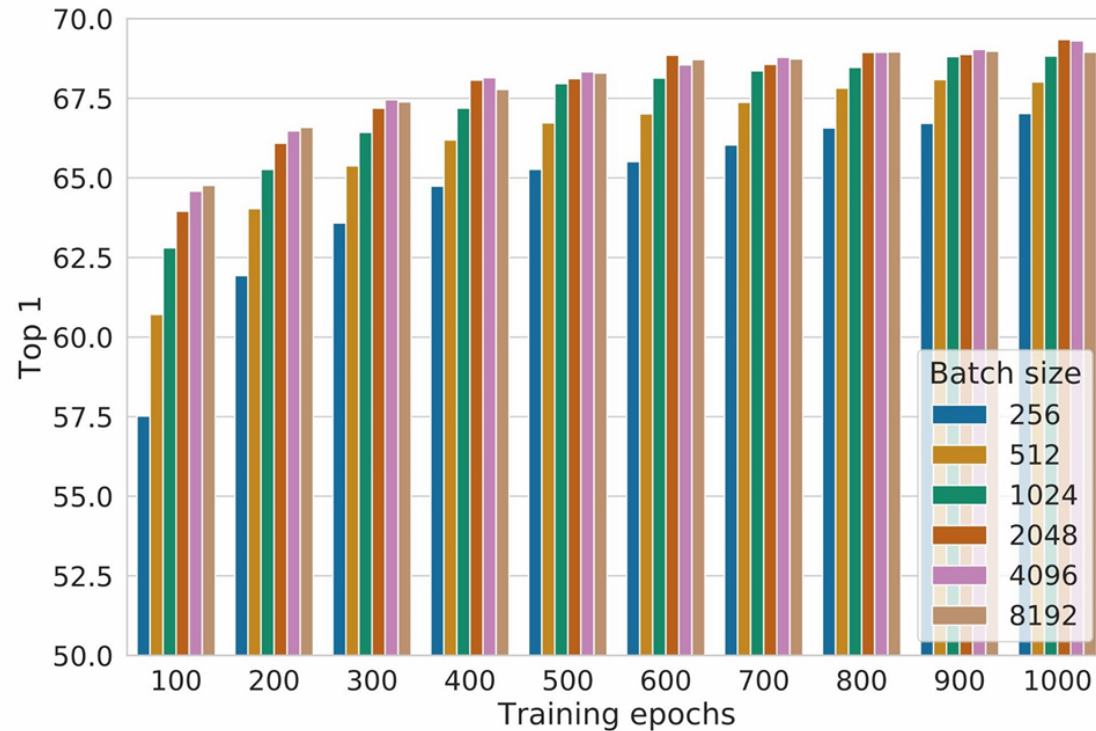


(i) Gaussian blur



(j) Sobel filtering

Large batch size is important for SimCLR



Large training batch size is crucial for SimCLR!

Large batch size causes large memory footprint during backpropagation:
requires distributed training on TPUs
(ImageNet experiments)

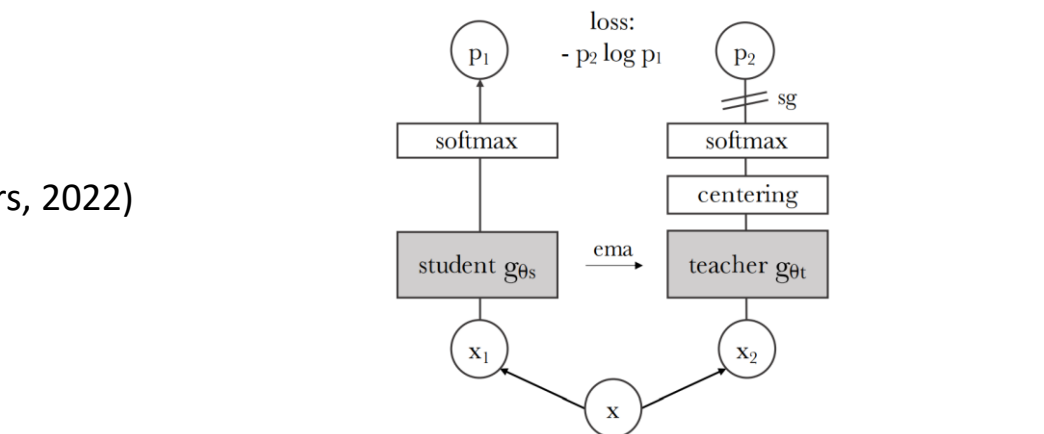
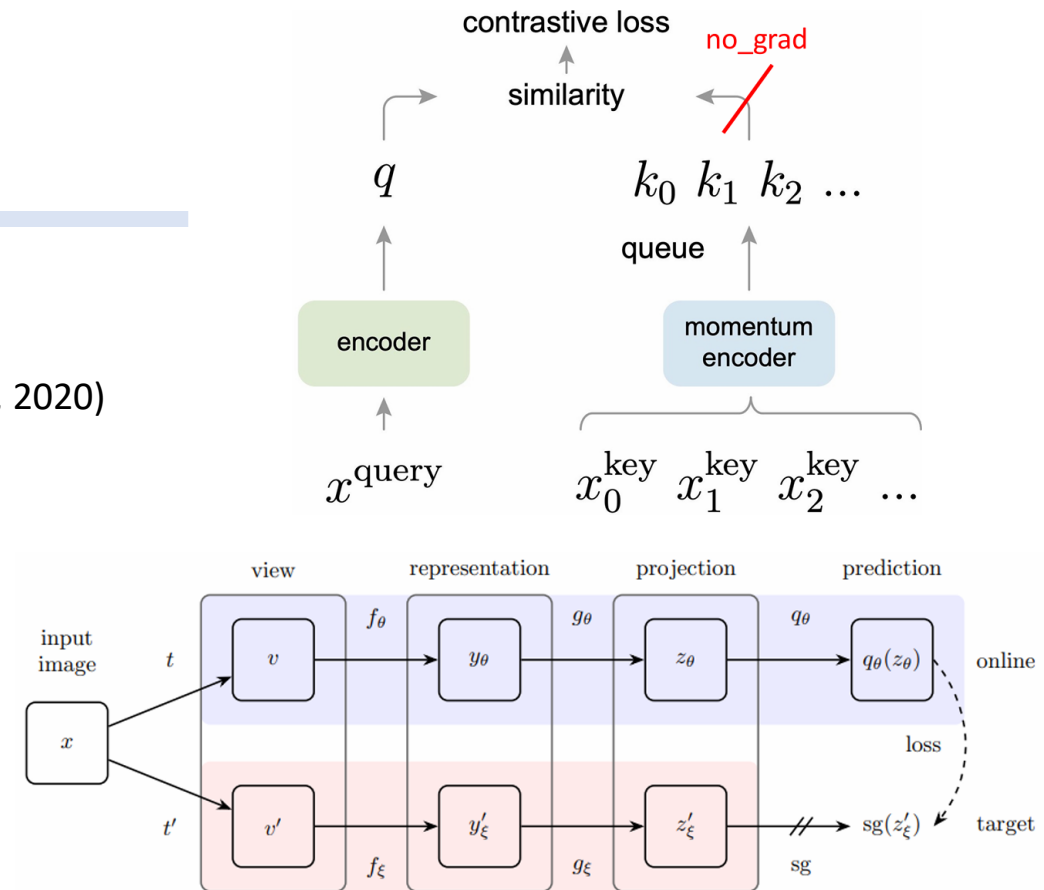
*Figure 9. Linear evaluation models (ResNet-50) trained with different batch size and epochs. Each bar is a single run from scratch.*¹⁰

Batch size problem

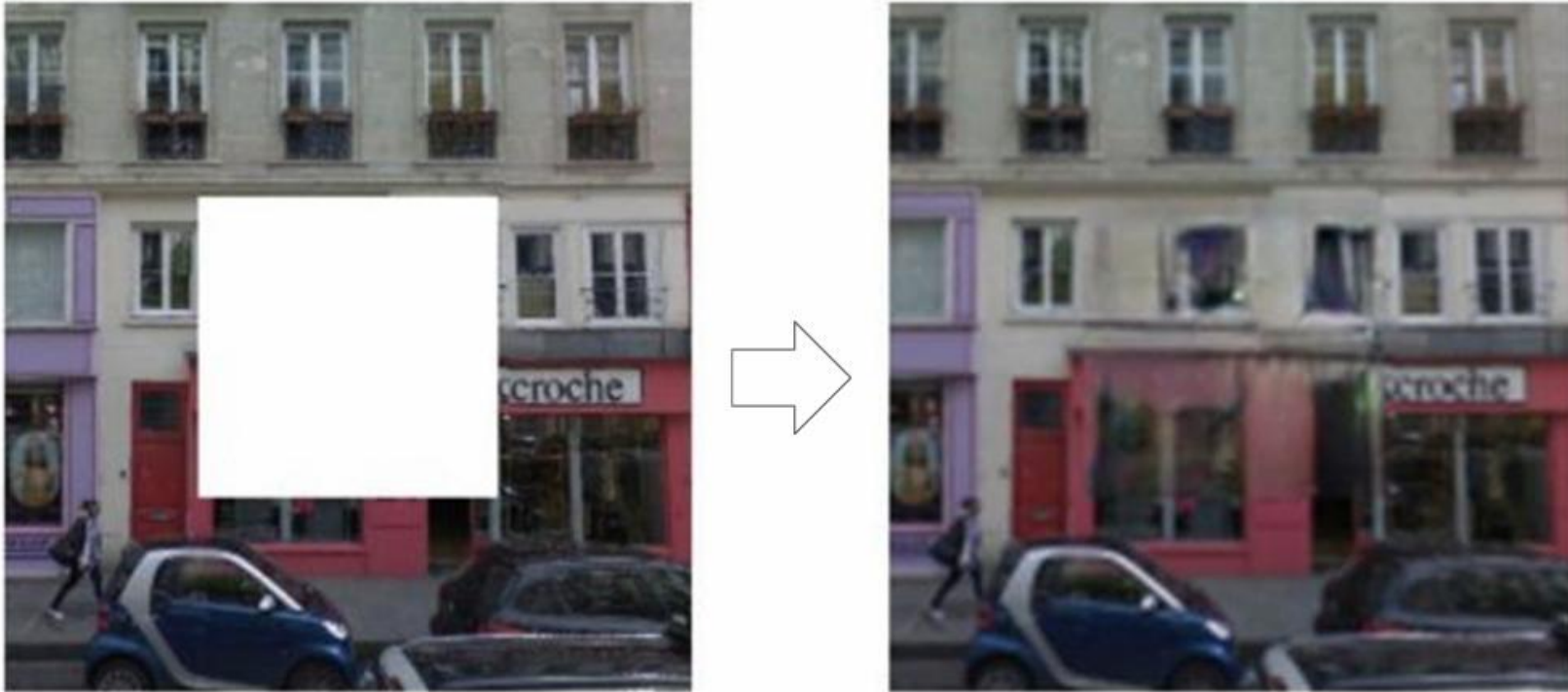
- Momentum Contrastive Learning (MoCo) v1, v2
(Chen et al., Improved Baselines with Momentum Contrastive Learning, 2020)

- BYOL: Bootstrap Your Own Latent
(J.B. Grill et al., Bootstrap Your Own Latent, 2020)

- DINO
(Caron et al., Emerging Properties in Self-Supervised Vision Transformers, 2022)

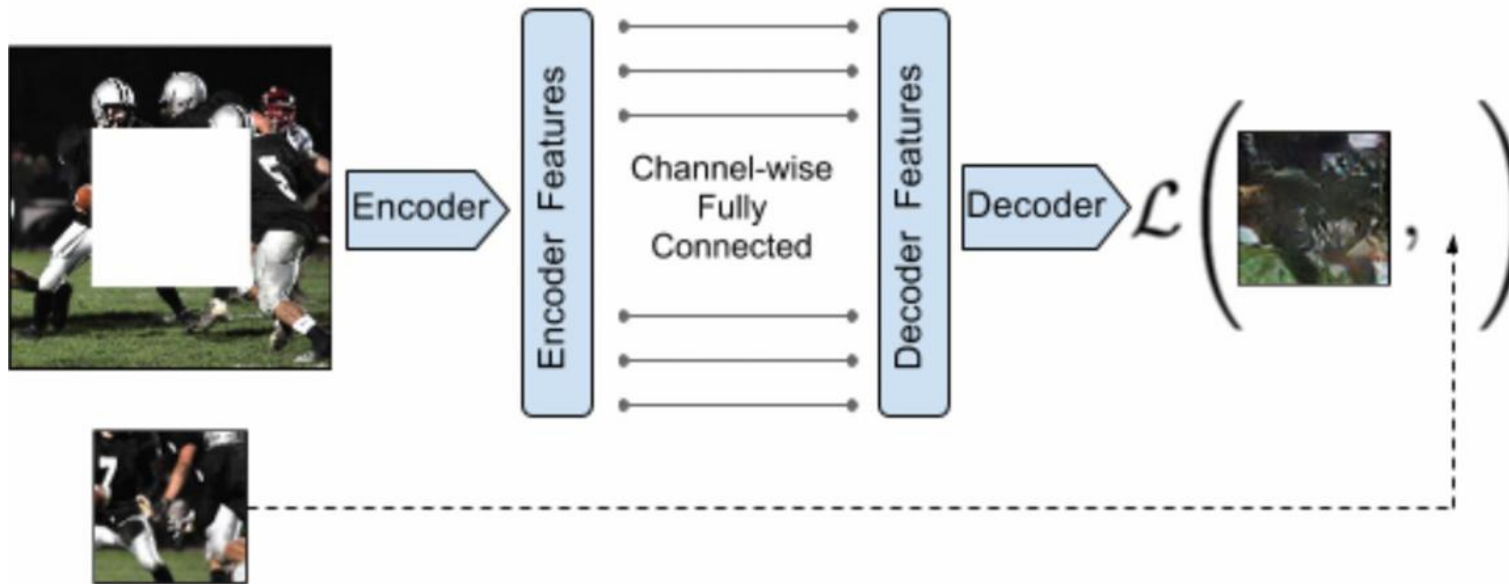


Pretext task: predict missing pixels (inpainting)



Context Encoders: Feature Learning by Inpainting (Pathak et al., 2016)

Learning to inpaint by reconstruction

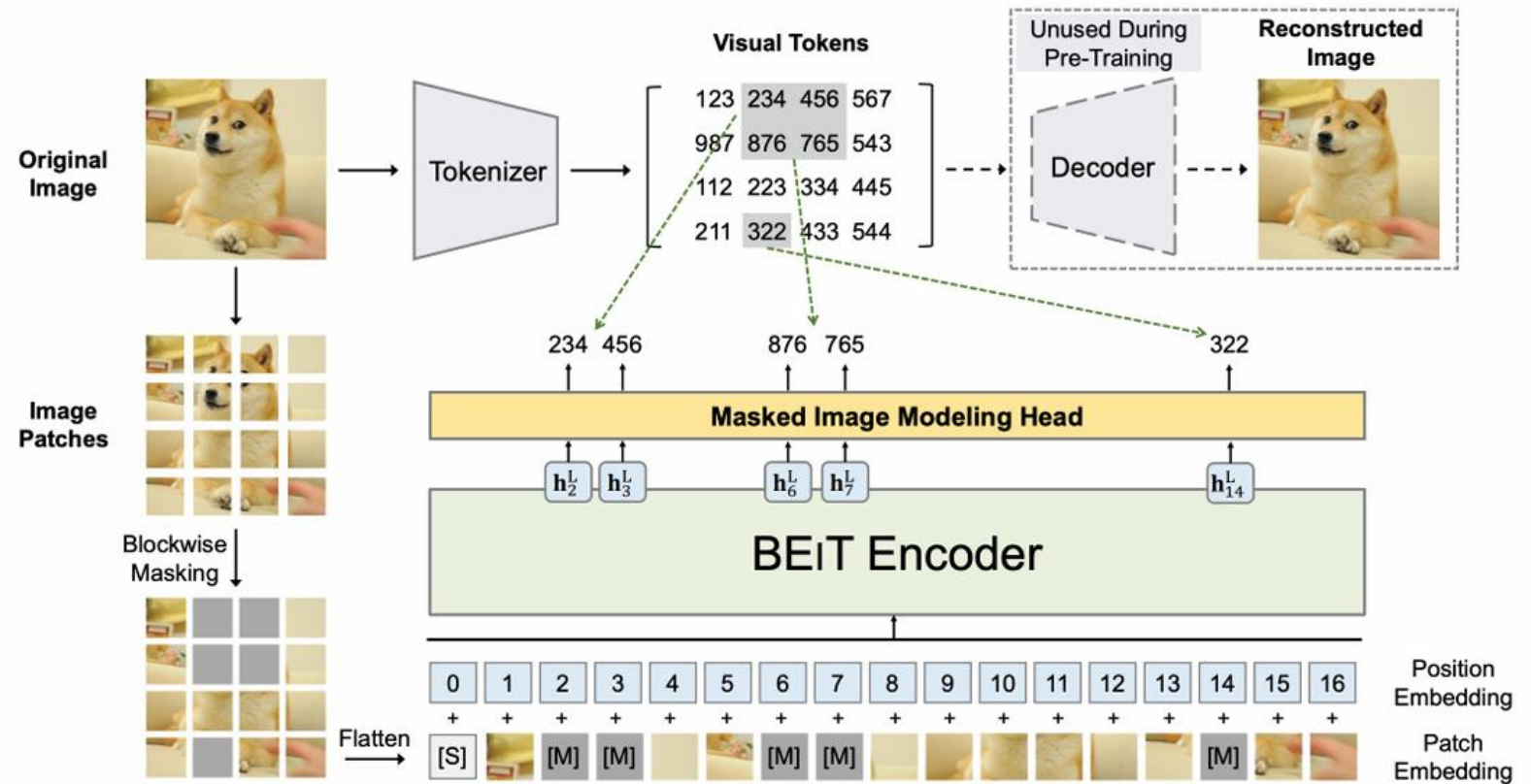


Learning to reconstruct the missing pixels

Masked Image Modeling (MIM)

- Bidirectional Encoder representation from Image Transformers

Breaks down the input image into discrete visual tokens, and then randomly masks a subset of the image patches



Masked Image Modeling (MIM)

- Masked Autoencoder (MAE)

treats the task as a regression problem with the original pixels as its target

